

ACTIVITY 9 — LEARNER WORKSHEET**Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen**

TYPE	DURATION	DMAIC PHASE
Core Activity	60 min	IMPROVE

Name	
Date	

What you are doing

Generate and select countermeasures that attack the root cause you proved — not the symptom. Apply the core Lean improvement tools: 5S, poka-yoke, visual management, standard work and Kaizen.

What you will produce: A prioritised countermeasure set traced to proven root causes, a 5S plan, one poka-yoke design and draft standard work.

Tools and techniques: Brainstorming, 5S, poka-yoke, visual management, standard work, Kaizen

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

templates/ — the worksheets you complete:

- 5s-plan.csv (S, what_it_means_here, action, owner)
- countermeasures.csv (root_cause, countermeasure, type_5S_pokayoke_standardwork, impact_H_M_L, effort_H_M_L, expected_effect)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Your steps

1. Generate countermeasures for each proven root cause — quantity before quality.
2. Prioritise using an impact vs effort matrix to find the quick wins.
3. Apply 5S thinking — Sort, Set in order, Shine, Standardise, Sustain — including to digital work.
4. Design one poka-yoke that makes the error difficult or impossible to make.
5. Draft standard work so the improved method is repeatable by anyone.
6. Plan a Kaizen event or pilot to test the countermeasure at small scale first.
7. Distinguish a containment countermeasure from a permanent solution.

Your working space

Complete every field. Keep this worksheet — it is your reference for applying the method at work.

Step 1 — Generate countermeasures for each proven root cause — quantity before quality.

Step 2 — Prioritise using an impact vs effort matrix to find the quick wins.

Step 3 — Apply 5S thinking — Sort, Set in order, Shine, Standardise, Sustain — including to di...

Step 4 — Design one poka-yoke that makes the error difficult or impossible to make.

Step 5 — Draft standard work so the improved method is repeatable by anyone.

Step 6 — Plan a Kaizen event or pilot to test the countermeasure at small scale first.

Step 7 — Distinguish a containment countermeasure from a permanent solution.

Answers to the discussion prompts

1. Does your poka-yoke PREVENT the error or merely DETECT it? Which is stronger?

2. Which countermeasure is a containment, and which is permanent?

3. What would you pilot first, and how would you know it worked?

Reflect & discuss

Where could a poka-yoke remove an error you personally make each week?

What standard work exists in your team but is not written down?

Check your work

Every countermeasure traces back to a proven root cause, and your poka-yoke prevents rather than detects the error.

Certified Lean Six Sigma Yellow Belt (CLSSYB) · C524 · Version v1 · © 2026 Tertiary Infotech Academy Pte Ltd